Spike-Driven Dreamer: Persistent Dendrites Give Spiking World Models Long-Term Memory

Anonymous CVPR submission

Paper ID *****

Abstract

Spiking neural networks (SNNs) compute with sparse binary spikes and run on neuromorphic hardware at a fraction of the energy of dense networks, which makes them a natural choice for embodied agents. Such agents see only part of their world, and a world model gives them memory by compressing past observations into a recurrent state. We find that existing spiking world models forget within a few steps, even when their neurons have long time constants. The cause is reset: membrane potentials are cleared at every environment step and dendrites at every somatic spike, so all memory has to pass through binary spikes at every step. Causal interventions on a spiking world model confirm this reset amnesia and show that long time constants also need a normalized write gain. Building on these findings, we propose the Spike-Driven Dreamer (SD-Dreamer), a spiking world model in which dendrites store memory and spikes communicate it. Its persistent dendritic neurons keep integrator and resonant dendritic channels that spikes never reset, with retention times from 1 to 1000 environment steps. The integrators hold what the agent has seen, and the resonators also record when. A stateless soma reads the channels and sends a spike count, the only signal that passes between neurons. On BSuite memory length, SD-Dreamer recalls a cue 101 steps back within the budget of R2I, while DreamerV3 fails beyond 51 steps. On POPGym RepeatPrevious, it is the only model, spiking or not, that learns to name the card shown exactly 32 steps earlier within 2M environment steps, and its world model also recalls exact lags of 64 steps on a task held out from development. The agent uses 4.6× less arithmetic energy per step than DreamerV3. Spiking world models thus remember over long horizons once their memory lives in dendrites that spikes do not reset.

1. Introduction

World models [10–12, 14] compress the history of observations and actions into a recurrent latent state, predict the future from it, and let an agent learn behavior in imagination. In partially observable environments, the agent must retain information over many steps, so this latent state serves as its memory. Memory benchmarks test this ability [26, 32, 33], and world models built on state space models retain information for hundreds of steps [7, 38]. However, these world models are artificial neural networks (ANNs) that compute with dense multiply-accumulate (MAC) operations at every environment step, which makes closed-loop inference costly for robots that run on batteries. Therefore, embodied agents need world models that keep a long memory at a low energy cost per step.

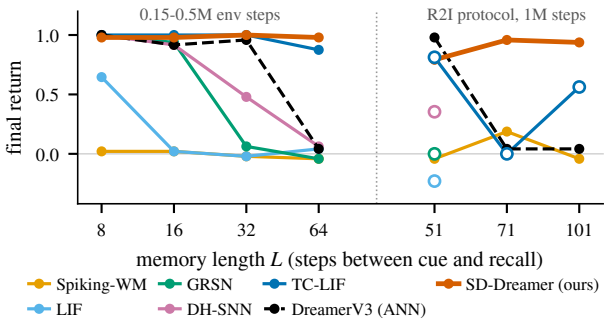


Figure 1. **Spiking world models forget.** Final return on BSuite memory length [32]: the agent sees a cue, then L uninformative steps, and is rewarded for recalling the cue (1 = always correct, 0 = chance). All models share one spiking world model and differ in the recurrent core (Sec. 5.1). Left: budgets of 150k to 500k environment steps. Right: L = 51, 71 and 101 with the lengths and budget of R2I [38] (1M steps). Spiking-WM fails at every length, recurrent LIF neurons stop at 8 steps, and GRSN and DH-SNN at 16; TC-LIF, whose state is carried across steps and never cleared, learns 51 and 101 steps (71: in progress). SD-Dreamer reaches a return of 0.9 at every length up to 101 steps, and DreamerV3 up to 51 (Tab. 2). Open markers: runs in progress.

Spiking neural networks (SNNs) [24] offer an energy-efficient alternative through sparse, event-driven computation [37]. Unlike ANNs, spiking neurons trigger synaptic

computation only when spikes arrive and replace MACs with accumulate (AC) operations, which take about a fifth of the energy of a MAC on a 45 nm chip [16]; neuromorphic processors exploit this sparsity [6, 30]. Moreover, spiking neurons are dynamical systems with membrane and dendritic time constants, and these dynamics are widely believed to provide temporal memory [3, 49, 51]. These properties make SNNs a natural basis for world models of the Dreamer family, which update a recurrent state at every step of closed-loop control and, in memory tasks, must keep information in this state over many steps.

Existing work on memory in SNNs designs richer neuron dynamics, such as adaptive thresholds [3, 47], dendritic branches with heterogeneous time constants [51], two compartments [49] and gated recurrence [35], and a recent spiking world model builds on multicompartment neurons whose dendrites integrate over long timescales [41]. These designs treat memory as a matter of longer time constants and overlook how the neuron state evolves across environment steps. Spiking reinforcement learning (RL) methods clear the membrane potentials at the start of every environment step [21, 41, 42], so the time constants act only within a single step. In the multicompartment neurons of spiking world models, a somatic spike also resets the dendrites [41], so a dendritic trace survives only until the neuron fires again. Moreover, the standard Euler update writes input with a gain of $1/\tau$, so compartments with long time constants store their input too weakly to drive spikes. As a result, existing spiking world models forget within a few steps (Fig. 1); we term this failure reset amnesia and confirm its causes with interventions (Sec. 3). Despite these advances in neuron dynamics, building a spiking world model that retains what it saw, and when, over long horizons remains challenging.

To address these problems, we propose the Spike-Driven Dreamer (SD-Dreamer), a spiking world model built from persistent dendritic neurons (PDNs) (Sec. 4). As shown in Fig. 2, a PDN removes each cause of reset amnesia. First, its linear dendritic channels persist across environment steps, so information outlasts a single step. Second, spikes write into these channels through synapses but never reset them, so a dendritic trace survives later spikes. Third, the write gain is normalized across retention times from 1 to 1000 environment steps, so channels with long retention times store their input at the same scale as short ones. In addition, integrator channels hold what was seen, and resonant channels, damped oscillators in the tradition of resonate-and-fire neurons [19], also hold when it was seen. A stateless soma reads the channels and sends a spike count, the only signal that other neurons receive; the graded state stays inside the neuron, as on neuromorphic cores [30]. We evaluate SD-Dreamer on BSuite memory length, POPGym RepeatPrevious and Memory Maze with their published protocols, and on DMC as a control task without a memory demand (Sec. 5). On BSuite, it reaches a return of 0.9 at every memory length up to 101 steps within the budget of R2I [38], where DreamerV3 stops at 51. On POPGym RepeatPrevious Medium [26], which rewards naming the card of exactly 32 steps ago, it is the only model, ANN baselines included, whose return rises above chance within 2M steps. World model probes show that only SD-Dreamer recalls exact lags of 32 and 64 steps, and only with its resonant channels; ablations confirm that this memory also requires the persistent channels and the normalized write gain. Its agent uses 4.6× less arithmetic energy per step than DreamerV3. The main contributions are as follows:

- We identify reset amnesia in spiking world models, caused by the step reset, the spike reset and a write gain tied to the time constant, and establish it with causal interventions.
- We propose the persistent dendritic neuron, in which persistent integrator and resonant dendrites with a normalized write gain store memory and a stateless soma communicates by spike counts.
- Building on this neuron, we develop SD-Dreamer and give the first evaluation of spiking world models on standard memory benchmarks against five spiking recurrent cores and DreamerV3, with DMC as a control and the energy per step.

2. Related Work

World models and memory. Latent world models learn a recurrent state from observations and train behavior in imagination [10–14]. Their memory is that of the recurrent core: a GRU in the Dreamer family, attention over a context window in transformer world models [25, 36, 50], and a state space recurrence in S4WM [7] and R2I [38]. R2I solves memory tasks that DreamerV3 cannot, and we use its settings for BSuite memory length [32] and POPGym [26]. Memory Maze [33] measures long-term memory in first-person 3D navigation with an offline probe of the world model state, in the spirit of probing the maps that navigation agents build [45]. Ni et al. [29] separate the length of memory from the length of credit assignment; our world model probes do the same by removing the policy. All these models are dense ANNs.

Memory cells for RL. Dense memory cells built on linear recurrences with decaying, rotating state include the LRU [31], Legendre memory units [44], S5 layers for in-context RL [23], and fast and forgetful memory, whose complex traces drift in phase with time [27]. They pass graded vectors between units; a PDN keeps such state inside each spiking neuron, without gates, and sends only spikes.

Spiking reinforcement learning. Spiking actors and Q-networks [21, 42] simulate a feedforward SNN for a few

steps per environment step and clear it before the next. Spiking-WM [41] brings multicompartment neurons into the world model of DreamerV3 and evaluates it on DeepMind Control from pixels and on speech classification. GRSN [35] aligns a gated recurrent spiking neuron with environment time and carries its state across steps for model-free RL in POMDPs. Spiking model-based control has also been trained end to end with recurrent LIF networks [17]. These works evaluate control and classification; on memory benchmarks, they remember at most 16 steps (Sec. 5.2).

Spiking neurons with long memory. Adaptive thresholds [3, 47], two-compartment neurons [49], heterogeneous dendritic time constants [51] and resonate-and-fire dynamics [15, 19, 48] extend the memory of spiking neurons. Removing the reset makes the membrane linear and parallelizable [9], and spiking state space layers attach spiking outputs to linear recurrences [2, 8, 39, 40]. These neurons are trained and tested on offline sequence classification, where the network sees the whole sequence and is reset between samples. A world model filters an unbounded stream step by step and also rolls its state forward in imagination; we run DH-SNN and TC-LIF in this setting as baselines. SD-Dreamer is designed for it: memory is carried across environment steps only in dendrites that no spike resets, retention times are counted in environment steps, and the soma that produces spikes holds no state. Biological dendrites act as computational subunits [4, 20, 22, 34] and many neurons resonate below threshold [18]; PDNs borrow both properties.

3. Reset Amnesia

3.1. Spiking world models

We follow the recurrent state space model of DreamerV3 [14]. At environment step t a recurrent core updates a deterministic state h_t from h_{t-1}, the previous stochastic latent z_{t-1} and action a_{t-1}; the posterior infers z_t from h_t and the observation embedding, the prior predicts it from h_t alone, and decoder, reward and continuation heads read (h_t, z_t). In a spiking world model [41] every layer is spiking and is simulated for T steps per environment step ($T = 5$), and $h_t \in \{0,1\}^{T\times N}$ is the spike output of the core.

Spiking-WM uses multicompartment neurons (MCNs). In simulation step k of environment step t, a basal and an apical dendrite integrate synaptic input with leaks discretized by the Euler method, and the soma integrates the basal potential gated by the apical one:

$$v^c[k] = v^c[k-1] + \tfrac{1}{\tau_c}\big(I^c[k] - v^c[k-1]\big), \quad c \in \{\text{bas}, \text{api}\},$$
$$s[k] = \mathrm{H}\big(v^{\text{som}}[k] - \vartheta\big), \tag{1}$$

where $I^c[k] = W_c\,[\,x_t[k];\, h_{t-1}[k]\,]$ combines the current input spikes with the core's spikes of the previous environment step and H is the Heaviside step, trained with a surrogate derivative [28, 46]. Two resets act on this neuron:

1. **Step reset.** All compartments are set to zero at the start of every environment step.
2. **Spike reset.** When the soma fires, the soma and both dendrites are set to zero.[1]

Consequences. With the step reset, the core computes $h_t = \Phi(x_t, h_{t-1})$ from a zero state: it is a binary recurrent network, and everything it knows about the past must be encoded again into the $T \times N$ bits of h_{t-1} at every step. The gradient from a reward L steps after a cue crosses L surrogate derivatives. The time constants τ_c act only within the T simulation steps of one environment step, which explains why changing them barely changes the performance of Spiking-WM [41]. The spike reset additionally truncates a dendritic trace at the next spike: with $\alpha = 1 - 1/\tau_c$,

$$\frac{\partial v^c[k]}{\partial I^c[k-j]} = \frac{1}{\tau_c}\,\alpha^j \prod_{i=1}^{j} \big(1 - s[k-i]\big), \tag{2}$$

which vanishes as soon as the neuron has fired once, so the effective memory is the shorter of τ_c and the interval between spikes.

Write gain. Removing both resets does not make long time constants useful by itself. The Euler update in Eq. (1) writes input with gain $1/\tau_c$, so for white input the stationary spread of v^c shrinks as $1/\sqrt{2\tau_c}$: a long time constant retains a cue but stores it faintly. With both resets removed (Sec. 3.2), a cue moved the basal potential of a $\tau_c = 128$ dendrite by 0.004 to 0.009, about 1% of the threshold $\vartheta = 0.5$, against 0.19 at $\tau_c = 2$.

3.2. Causal test

We test these mechanisms with interventions on Spiking-WM in a world model probe that isolates memory from exploration and credit assignment. The data are BSuite memory length episodes [32] collected by a random policy, with the terminal reward set to the cue (± 1, cue recall); only the world model is trained, and we measure the AUC of its reward prediction at the recall step on evaluation episodes, from 0.5 without memory to 1 with perfect recall (details in Supp. C). Three interventions target the mechanisms above: **A** keeps all compartments across environment steps and clears them only at episode starts; **B** keeps the dendrites when the soma fires; **G** multiplies the dendritic input by $\sqrt{\tau_d/2}$, which equalizes the stationary spread across dendritic time constants τ_d. Table 1 gives four findings.

[1] The equations of [41] show the somatic reset; the dendritic reset is part of its official implementation (`MCNode.calc_spike`), which all our Spiking-WM runs use.

Table 1. **Interventions on Spiking-WM** (world model probe, cue recall over L steps). Updates until the reward AUC exceeds 0.9, two seeds; – = not within 12k updates; blank = not run. A: state kept across environment steps; B: no dendritic reset on a somatic spike; G: dendritic input scaled by $\sqrt{\tau_d/2}$. GRU: reference core with continuous state.

	A	B	G	τ_d	$L=8$	$L=16$	$L=32$
Spiking-WM				2	– / –	– / –	12k / –
				16	– / –	– / –	
				128	– / –	– / –	
B only		✓		128	– / –		
G only			✓	128		– / –	
A	✓			2	– / 9k	– / –	
	✓			128	3k / 4k	8k / 8.5k	
A+B	✓	✓		128	3k / 3k	4.5k / 4.5k	– / –
A+B+G	✓	✓	✓	128	**0.5k / 0.5k**	**0.5k / 0.5k**	**2k / 3k**
GRU core	continuous state				0.5k / 0.5k	1k / 1k	5k / 3.5k

With resets, time constants do not matter. Spiking-WM fails at 8 and 16 steps for $\tau_d \in \{2, 16, 128\}$, and the cue can be decoded from its spikes for only 2 to 3 steps after it is shown. At 32 steps, one of two seeds crosses an AUC of 0.9 at the last of 12k updates by holding the cue in self-sustained recurrent firing, a memory that exists only as spikes.

Persistence is necessary. Without A, neither B nor G helps. With A and $\tau_d = 128$, the world model recalls 8 and 16 steps; removing the dendritic reset too (A+B) nearly halves the updates at 16 steps (4.5k instead of 8k to 8.5k).

The write gain limits long time constants. A+B fails at 32 steps although a linear probe still decodes the cue from the membrane potentials at the recall step (accuracy 0.88): the trace is there, but too faint to drive spikes. Restoring the gain (A+B+G) recalls 8 and 16 steps within 500 updates and 32 steps within 2000 to 3000, faster than the GRU core. The gain does not act through the firing rate: capped at a rate of 0.06, A+B+G still learns 32 steps in 2000 updates.

Longer memory needs more than removing resets. A+B+G fails at 64 steps even with $\tau_d = 512$, and in reinforcement learning it stops at 32 steps (Spiking-WM A+B+G in Supp. A). Its time constants are counted in simulation steps, and its soma still resets. Section 4 removes both; in our model, a soma with a membrane and a reset loses the recall of 64 steps (Sec. 5.5).

4. Spike-Driven Dreamer

Section 3 gives a design rule: memory must live in a state that spikes never reset, written with a gain that does not vanish for long time constants. We build SD-Dreamer from neurons that follow this rule, keeping the rest of the spiking world model unchanged (Fig. 2).

4.1. Persistent dendritic neuron

A layer has N neurons with M dendritic channels each. At environment step t it receives the firing rate $\boldsymbol{x}_t$ of its input spikes over the T simulation steps and its own firing rate $\bar{\boldsymbol{s}}_{t-1}$ of the previous step. The synaptic current into the dendrites is

$$\boldsymbol{w}_t = \mathrm{LN}\big(B\boldsymbol{x}_t + R\,\bar{\boldsymbol{s}}_{t-1}\big) \in \mathbb{R}^{NM}, \tag{3}$$

where B and R are synaptic matrices applied to spikes and LN is layer normalization [1]. Each channel has a retention time τ, in environment steps, and a decay $\lambda = e^{-1/\tau}$.

Integrator channels add the current to a leaky trace,

$$d_t = \lambda\, d_{t-1} + \sqrt{1-\lambda^2}\; w_t. \tag{4}$$

Resonant channels are damped oscillators with two state variables that rotate by a phase θ at every step:

$$\begin{pmatrix} a_t \\ b_t \end{pmatrix} = \lambda \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \begin{pmatrix} a_{t-1} \\ b_{t-1} \end{pmatrix} + \sqrt{1-\lambda^2} \begin{pmatrix} w_t \\ 0 \end{pmatrix}. \tag{5}$$

An integrator channel is the special case $\theta = 0$ with $b \equiv 0$.

The soma has no state. It reads the channels of its neuron and the direct input,

$$u_t = \textstyle\sum_{m=1}^{M} \big(C_m a_{t,m} + C'_m b_{t,m}\big) + D\boldsymbol{x}_t, \tag{6}$$

where C_m, C'_m are weights of each neuron and D is a synaptic matrix, and in simulation step k it emits $s_t^k = \mathrm{H}\big(\mathrm{N}(u_t) - k\vartheta/T\big)$, $k = 1, \ldots, T$, with N the normalization used throughout the world model and $\vartheta = 0.5$. The number of spikes in an environment step thus encodes u_t in $T+1$ levels. These binary frames are the only signal the neuron sends; the next layer and the feedback read their rate $\bar{s}_t$.

Parameters. We use $N = 512$ and $M = 2$: one integrator and one resonant channel per neuron. Retention times are initialized uniformly on a log scale in $[1, 1000]$ environment steps and phases uniformly in $[0, 2\pi)$ (the initialization of the LRU [31]); both are learned. The Heaviside step is trained with a surrogate derivative, and all states are reset only at the start of an episode.

4.2. Why the neuron remembers

No reset in the memory path. Along the decay of a channel, the influence of a current k steps back is $\partial d_t / \partial w_{t-k} = \sqrt{1-\lambda^2}\,\lambda^k$ for an integrator and $\sqrt{1-\lambda^2}\,\lambda^k R(k\theta)$ for a resonator, whatever the neuron fired in between. This path

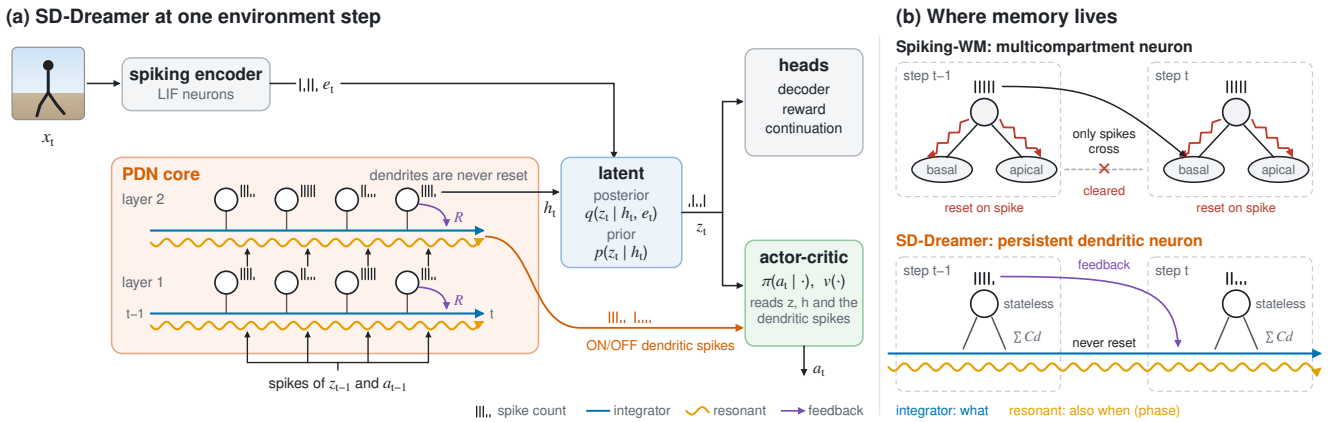


Figure 2. **Spike-Driven Dreamer.** (a) Two layers of persistent dendritic neurons (PDNs) form the recurrent core of a spiking Dreamer. Each neuron has an integrator (blue) and a resonant (orange) dendritic channel, which carry the memory across environment steps and are never reset; its stateless soma sends a spike count over the $T = 5$ simulation steps, and the spikes of a layer also feed back into its dendrites (R). The actor and critic also read the dendrites of the second layer as ON and OFF spikes. (b) In Spiking-WM, all compartments are cleared at every step and the dendrites at every somatic spike, so only spikes reach the next step. In a PDN, the dendritic channels persist; spikes add to them but never clear them.

never crosses a surrogate derivative, so the memory and its gradient decay with the retention time alone. The feedback $R\bar{s}_{t-1}$ adds a second path, through the threshold, that writes into the channels and never clears them; the memory holds without it, as the world model without feedback still recalls 64 steps and the exact lag of 32 steps (Sec. 5.5).

Normalized write. For white input of unit variance, Eq. (4) gives $\mathrm{Var}(d_t) = (1-\lambda^2)\sum_{k\geq 0}\lambda^{2k} = 1$ for every τ, whereas the Euler gain $1-\lambda$ of Eq. (1) gives $(1-\lambda)/(1+\lambda) \approx 1/(2\tau)$. The LRU normalization [31] thus keeps channels with retention times from 1 to 1000 steps on one scale; intervention G is the same scaling up to a factor of 2.

A spectrum of retention times. Retention times spread uniformly on a log scale cover memory lengths from one to a thousand steps, so no time constant is tuned per task.

Resonance tells time. An input that arrived k steps ago contributes $\lambda^k(\cos k\theta,\ \sin k\theta)\, w_{t-k}$ to a resonant channel: its phase identifies the lag. In integrator channels, inputs of different ages are superposed with smoothly decaying weights λ^j, so isolating the input of exactly k steps ago from those of $k \pm 1$ steps ago requires subtracting channels with nearby retention times, a poorly conditioned readout. Integrators hold what was seen, and resonators also hold when it was seen.

Spike counts and feedback predict motion. One spike per step gives a downstream neuron one bit, a spike count $T+1$ levels; the feedback adds a nonlinear recurrence that linear dendrites lack. On the same pixel walker episodes, the world model with both predicts the next frame as well as recurrent LIF neurons (image loss 44.3 vs. 44.1), and with one spike per step it does not (55.3; Supp. F).

Spikes communicate. All signals between neurons are spikes, so the synaptic operations $B\boldsymbol{x}_t$, $R\bar{\boldsymbol{s}}_{t-1}$ and $D\boldsymbol{x}_t$ are accumulates. The dendritic recurrence and the readout are local to a neuron, and the normalizations need only the mean and variance of a layer, as in every spiking world model we compare: $O(NM)$ multiply-accumulates per step, against $O(N^2)$ for the dense recurrent matrix of a GRU.

4.3. Spiking world model

SD-Dreamer replaces the recurrent core of the spiking world model (Sec. 3.1) by two PDN layers; the second layer reads the spikes of the first, and its spikes form the deterministic state h_t. Everything else is unchanged and spiking: the input layer that reads (z_{t-1}, a_{t-1}), the encoder, the posterior and prior over z_t (32 categorical variables with 32 classes), and the decoder, reward and continuation heads, all built from LIF neurons with $T = 5$ simulation steps, trained with the losses of DreamerV3 [14]. The dendritic state is carried through training sequences and through imagination rollouts, and is cleared only at episode starts.

Policy input. The actor and critic read z_t, h_t and the state of the dendrites of the second layer, on every task; the Memory Maze probes read the same features. The soma sends at most T spikes per neuron and step, whereas the dendrites hold the memory in graded form. To pass them on as spikes, each state variable is standardized over the layer and encoded

by T ON and T OFF thresholds at $\pm(k-\frac{1}{2})\frac{2}{T}$, $k = 1, \ldots, T$, giving $2T$ binary frames whose rates represent $2T + 1$ levels. With three state variables per neuron (the integrator and the two of the resonator), the actor and critic receive 3072 additional binary inputs per simulation step, the only path by which the dendritic state leaves a neuron; their first layer reads these inputs through a factor of rank 64. Supp. E compares a policy that reads (z_t, h_t) only.

5. Experiments

5.1. Setup

Models. All models share one spiking world model, the network of Spiking-WM [41] built on DreamerV3 at the size of DreamerV3-S (512 units, $T = 5$), and differ in the recurrent core: **Spiking-WM** (multicompartment neurons with both resets, $\tau = 2$); **recurrent LIF** neurons, cleared at every environment step as usual in spiking RL; **GRSN** [35], gated recurrent spiking neurons; **DH-SNN** [51], LIF neurons with eight dendritic branches of learned time constants; **TC-LIF** [49], two-compartment LIF neurons for long sequences (GRSN, DH-SNN and TC-LIF update once per environment step and carry their state); **DreamerV3** [14], the same architecture without spikes (GRU core, SiLU, one simulation step); and **SD-Dreamer** with two PDN layers. R2I [38], a DreamerV3 agent with a state space recurrence and the state of the art among ANN world models on memory tasks, runs with its official code and configuration on POPGym; on BSuite we compare with its published results. Within a benchmark, all models use identical hyperparameters, budgets and evaluation. The actor and critic of every model read (z_t, h_t); those of SD-Dreamer also read its dendritic state as spikes (Sec. 4.3), with 4.1M instead of 3.4M parameters (sizes in Supp. A, an actor-critic of 3.4M in Supp. E).

Benchmarks. We use benchmarks with published protocols and ANN results. **BSuite memory length** [32] shows a cue, then L uninformative steps, and rewards the agent for recalling the cue; we run $L \in \{8, 16, 32, 64\}$, and $L \in \{51, 71, 101\}$ with the lengths, budget (1M steps) and training windows of R2I [38]. **POPGym RepeatPrevious** [26] rewards the agent at every step for naming the suit of the card shown k steps earlier ($k = 4$ Easy, $k = 32$ Medium), one of the three POPGym tasks with the highest memory demand that R2I evaluates; we use the environment settings of R2I's code for 2M steps. **Memory Maze 9×9** [33] is first-person 3D navigation with episodes of 1000 steps; we use its offline probing benchmark with the official dataset and probes: after 1M world model updates with truncated backpropagation through time, probes decode the positions of the target objects (MSE) and the maze layout (accuracy) from the frozen state. **DMC** [43] from proprioception (walker walk, cheetah run, cartpole swingup; 500k steps, train ratio 512, as in DreamerV3) and walker walk from pixels (1M frames) serve as control tasks without a memory demand. On BSuite the reward depends on both the cue and the action, which the scalar reward head of a Dreamer world model does not learn within our budgets; all models therefore use a reward head with one output per action (Supp. B). Supp. A gives the settings and the evaluation protocol of each benchmark.

World model probes. To measure memory without exploration and credit assignment, we also train world models alone on data of a random policy and measure the AUC of their reward prediction where the reward depends on the past (Sec. 3.2): cue recall, where the terminal reward is the cue L steps back, and RepeatPrevious, where the reward indicates whether the action names the card k steps back. As in probing benchmarks [33], we report the final AUC after a fixed number of updates (Supp. C).

5.2. Memory in reinforcement learning

Table 2, Fig. 1 and Fig. 3 show how far each model remembers. The spiking baselines remember as far as their neurons keep their state. Spiking-WM, cleared at every step and at every spike, recalls no cue, not even 8 steps back, and recurrent LIF neurons, cleared at every step, solve 8 steps. GRSN and DH-SNN carry leaky states across steps and solve 16 steps. TC-LIF adds its input to two compartments that do not decay, and its spikes subtract a fixed amount from them instead of clearing them; it solves 64 steps and, with R2I's lengths, 51 and 101 steps (71: in progress), as the design rule of Sec. 3 predicts. DreamerV3 solves up to 32 steps in the short budgets and 51 steps with R2I's lengths (after 902k of 1M steps), and fails at 64, 71 and 101. **[TODO: Recurrent LIF, GRSN and DH-SNN at 51 steps (runs in progress).]**

SD-Dreamer reaches a return of 0.9 at every length, as fast as the fastest baseline or faster: 64 steps after 82k environment steps (TC-LIF 112k), and with R2I's lengths 51, 71 and 101 steps after 62k, 182k and 142k (TC-LIF 102k, >222k and 242k; returns in Supp. A). R2I reports a median return of about 0.9 at 51 and 71 memory steps and about 0.35 at 101 after 1M steps (10 seeds; Fig. 7 of [38]), with DreamerV3's default reward head and lengths that count one step less.

On RepeatPrevious Easy, SD-Dreamer reaches 0.72, close to the best baseline, recurrent LIF (0.74). On Medium, which asks for the card of exactly 32 steps ago, every baseline, DreamerV3 included, stays at chance after 2M steps, while SD-Dreamer reaches 0.48. R2I, run with its official code, stays at chance on both tasks within 2M steps. In its published curves, R2I leaves chance on Medium only after about 3M steps, and DreamerV3 learns Easy but not Medium (Fig. 9 of [38]), as in our runs. **[TODO: DH-SNN and TC-LIF on POPGym (runs in progress).]**

Table 2. **Memory in reinforcement learning.** BSuite memory length [32]: success as in R2I [38], i.e. whether the evaluation return reaches 0.9 within the budget (✓) or not (✗); steps and returns in Supp. A. †Lengths, budget and training windows of R2I (1M steps; reward head with one output per action). POPGym RepeatPrevious [26] with the environment settings of R2I's code: max-mean episodic return, the metric of POPGym (chance = −0.5). n/a: not run, as the model fails at a shorter length. ‡R2I with its official code and configuration for 2M steps, measured on its training episodes (for our models the two measures differ by at most 0.07). Blue: in progress.

	BSuite memory length: success							RepeatPrevious: return ↑	
	L=8	16	32	64	51†	71†	101†	Easy	Medium
Spiking-WM [41]	✗	✗	✗	✗	✗	✗	✗	0.64	-0.46
Recurrent LIF	✓	✗	✗	✗	·	n/a	n/a	0.74	-0.44
GRSN [35]	✓	✓	✗	✗	·	n/a	n/a	0.37	-0.47
DH-SNN [51]	✓	✓	✗	✗	·	n/a	n/a	-0.44	-0.46
TC-LIF [49]	✓	✓	✓	✓	✓	·	✓	-0.28	-0.48
DreamerV3 (ANN) [14]	✓	✓	✓	✗	✓	✗	✗	0.42	-0.41
R2I (ANN) [38]	reported in [38], see text							-0.48‡	-0.46‡
SD-Dreamer (ours)	✓	✓	✓	✓	✓	✓	✓	0.72	0.48
budget (env. steps)	150k	200k	300k	500k	1M	1M	1M	2M	2M

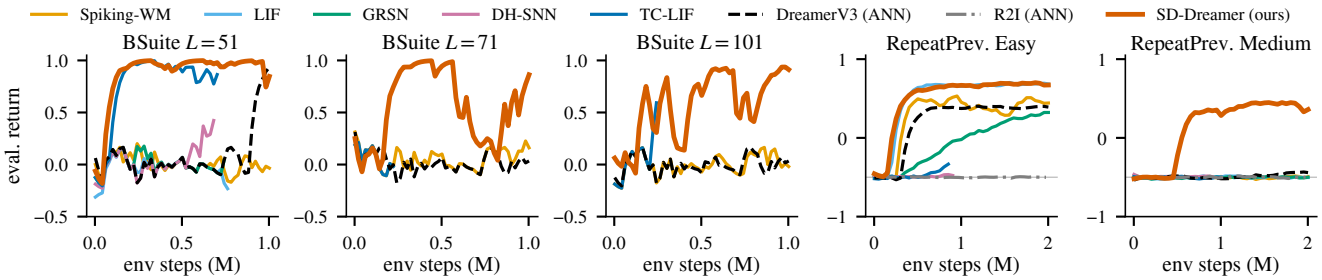


Figure 3. **Learning curves** on BSuite memory length with the lengths and budget of R2I (left three) and POPGym RepeatPrevious (right two; chance = −0.5). Exponential moving average of the evaluation return (R2I: return of the training episodes, see Tab. 2). Runs in progress.

Table 3. **World model probes** (data of a random policy, no policy). Final reward AUC (0.5 = no memory), mean of the last three evaluations after 12k (cue recall) or 10k (RepeatPrevious) updates; updates until 0.9 in Supp. C. *Held out (used for no design choice): 20k updates, training window 128. Blue: in progress.

	cue recall				RepeatPrevious		
	L=8	16	32	64	k=4	32	64*
Spiking-WM [41]	0.49	0.54	0.66	0.52	0.97	0.50	0.50
Recurrent LIF	1.00	0.52	0.50	0.50	0.99	0.50	0.50
GRSN [35]	1.00	1.00	0.51	0.51	0.89	0.51	0.50
DH-SNN [51]	1.00	1.00	1.00	0.52	0.63	0.50	0.50
TC-LIF [49]	1.00	1.00	1.00	1.00	0.86	0.50	0.57
DreamerV3 (ANN) [14]	1.00	1.00	1.00	0.50	0.95	0.50	0.50
SD-Dreamer (ours)	1.00	1.00	1.00	1.00	0.98	0.92	0.94

5.3. What the world model remembers

The probes separate memory from policy learning (Tab. 3). On cue recall, SD-Dreamer and TC-LIF recall 64 steps (AUC 1.00); DH-SNN and DreamerV3 stop at 32, GRSN at 16, recurrent LIF at 8, and Spiking-WM recalls no length. The order is that of RL, so the limits of the baselines are limits of their world models.

RepeatPrevious asks for a different memory: the input of an exact lag among a stream of similar inputs. At $k = 4$ most models learn it (AUC 0.86 to 0.99; SD-Dreamer 0.98). At $k = 32$, SD-Dreamer is the only model that learns it (AUC 0.92); all others, TC-LIF and DreamerV3 included, stay at 0.50 to 0.51. Resonance is the reason: without resonant channels the same model reaches only 0.57 (Tab. 5). TC-LIF recalls the cue 64 steps back as well as SD-Dreamer, but its compartments oscillate without decay at one period of about 12 steps that all neurons share (Supp. A), whereas each resonant channel of SD-Dreamer has its own phase and retention time. We added resonant channels after an earlier model with integrator channels only failed this task (Supp. F) and used RepeatPrevious Hard ($k = 64$) for no design choice. On Hard, SD-Dreamer reaches an AUC of 0.94, and all other models stay at 0.50 to 0.57.

5.4. Memory Maze

Memory Maze tests whether a world model builds a map of a 3D maze from first-person views over 1000 steps. Table 4

Table 4. **Memory Maze and DMC.** Memory Maze 9×9 offline probing [33]: probes on the frozen state after 1M updates, second half of the evaluation episodes (constant predictor: 23.9 MSE, 80.8% walls). DMC from proprioception: return after 500k steps; walker walk from pixels: return after 1M frames. n/a: not run. Bottom: reference results from [33] (larger RSSM).

	Memory Maze 9×9		DMC, proprio. ↑			pixels ↑
	obj. MSE ↓	walls % ↑	walk	run	swing	walk
Spiking-WM [41]	11.9	83.7	923	343	497	917
Recurrent LIF	11.7	83.7	807	532	811	478
GRSN [35]	11.5	83.3	863	392	652	389
DH-SNN [51]	n/a	n/a	n/a	n/a	n/a	417
TC-LIF [49]	n/a	n/a	n/a	n/a	n/a	457
DreamerV3 (ANN) [14]	9.8	85.3	953	699	871	968
SD-Dreamer (ours)	9.4	82.8	923	487	676	842
RSSM [33]	8.3	89.0	n/a	n/a	n/a	n/a
RSSM, TBTT [33]	6.2	94.0	n/a	n/a	n/a	n/a

Table 5. **Ablations** on the world model probes (as in Tab. 3). Each row changes one design choice of SD-Dreamer.

	cue recall		RepeatPrev.	firing
variant	L=32	64	k=32	rate
SD-Dreamer	1.00	1.00	0.92	0.25
dendrites cleared every step	0.51	0.50	0.50	0.25
single $\tau = 16$	1.00	0.49	0.92	0.24
single $\tau = 128$	1.00	1.00	0.93	0.26
τ fixed at initialization	1.00	1.00	0.93	0.24
Euler write gain $1 - \lambda$	1.00	0.52	0.76	0.21
one PDN layer	1.00	0.50	0.91	0.21
one PDN layer, $T = 1$	0.99	0.99	0.50	0.15
no resonant channels	1.00	1.00	0.57	0.27
all channels resonant	1.00	1.00	0.95	0.23
firing rate capped at 0.06	1.00	0.51	0.92	0.06
soma with membrane and reset	1.00	0.52	0.77	0.18
Heaviside soma	1.00	1.00	0.88	0.16
$M = 4$ channels	1.00	1.00	0.93	0.25
no spike feedback	1.00	1.00	0.94	0.23

reports the probes on the state that the policy reads: (z_t, h_t) for all models, and in addition the dendritic spikes for SD-Dreamer. SD-Dreamer locates the target objects as well as DreamerV3 (MSE 9.8) and better than every spiking baseline (11.5 to 11.8). On the wall layout, all models of this size stay within 5 points of the constant predictor (80.8%), SD-Dreamer at 82.8% and DreamerV3 at 85.3%; the larger RSSM of [33] reaches 89.0%. **[TODO: Memory Maze: final probes after 1M updates (in progress); the values are the latest probes during training.]**

5.5. Ablations

Table 5 changes one design choice at a time (updates until 0.9 in Supp. C). **Persistence**: clearing the dendrites at every environment step erases the memory (AUC 0.50 to 0.51 on all three tasks), the direct test of reset amnesia inside our own model. **Retention time**: a single retention time of 16 steps fails at 64 steps and one of 128 succeeds; the spectrum from 1 to 1000 steps succeeds without being set for the task, learned or kept at its initialization. **Write gain**: with the Euler gain $1 - \lambda$ the channels with long retention times are too faint, and the model fails at 64 steps (0.52) and reaches only 0.76 at $k = 32$. **Stateless soma**: a soma that keeps a membrane potential and is reset by its spike fails at 64 steps (0.52) and reaches 0.77 at $k = 32$. **Resonance**: without resonant channels the model fails at $k = 32$ (0.57); with resonant channels only it solves all three tasks, as does SD-Dreamer with one channel of each kind. **Depth and sparsity**: one PDN layer, or a firing rate capped at 0.06 (from 0.25), fails at 64 steps. The spike count, the feedback and the number of channels leave the memory nearly unchanged (AUC 0.88 to 0.94 at $k = 32$); the spike count and the feedback serve the prediction of motion (Sec. 4.2).

5.6. Control tasks and energy

On DMC from proprioception (Tab. 4), SD-Dreamer matches the best spiking model on walker walk (923, as Spiking-WM) and reaches 487 on cheetah run and 676 on cartpole swingup, where recurrent LIF reaches 532 and 811. An actor that reads (z_t, h_t) only reaches the same return with the same world model on walker walk and cheetah run (904 and 473) and a higher one on cartpole swingup (813; Supp. E). **[TODO: Walker walk from pixels after 1M frames (runs in progress).]**

Per environment step (Supp. H), the agent of SD-Dreamer, world model filter and actor, costs 4.7 μJ, 4.6× less than DreamerV3 (22.0 μJ); the filter alone costs 3.4 against 17.2 μJ. The spiking baselines cost 2.4 to 3.4 μJ, less than SD-Dreamer: its filter performs more accumulates (3.1M against 1.6 to 2.2M, at a recurrent firing rate of 0.18), and its actor also reads the 3072 dendritic spike inputs (1.3 against 0.3 to 0.6 μJ). As in SNN energy reports, the estimates count 32-bit arithmetic [16] and leave out memory access, which can dominate on hardware [5], for all models.

6. Conclusion

Existing spiking world models forget because their neurons are cleared at every environment step and at every spike, which leaves their time constants nothing to act on. SD-Dreamer keeps memory in persistent integrator and resonant dendrites, written with a normalized gain and never reset by spikes, and communicates through the spike counts of a stateless soma. It recalls cues 101 steps back on BSuite within the budget of R2I and is the only model in our comparison that recalls exact lags of 32 and 64 steps, at 4.6× less arithmetic energy per step than DreamerV3.

Limitations. The dendritic state is graded, so deployment needs neuromorphic cores whose neurons hold several state

variables and that compute layer normalization; our energy estimate counts arithmetic and leaves out memory access.

References

[1] Jimmy Lei Ba, Jamie Ryan Kiros, and Geoffrey E. Hinton. Layer normalization. *arXiv preprint arXiv:1607.06450*, 2016. 4

[2] Malyaban Bal and Abhronil Sengupta. P-SpikeSSM: Harnessing probabilistic spiking state space models for long-range dependency tasks. In *International Conference on Learning Representations (ICLR)*, 2025. 3

[3] Guillaume Bellec, Darjan Salaj, Anand Subramoney, Robert Legenstein, and Wolfgang Maass. Long short-term memory and learning-to-learn in networks of spiking neurons. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 2, 3

[4] David Beniaguev, Idan Segev, and Michael London. Single cortical neurons as deep artificial neural networks. *Neuron*, 109(17):2727–2739, 2021. 3

[5] Manon Dampfhoffer, Thomas Mesquida, Alexandre Valentian, and Lorena Anghel. Are SNNs really more energy-efficient than ANNs? an in-depth hardware-aware study. *IEEE Transactions on Emerging Topics in Computational Intelligence*, 7(3):731–741, 2023. 8

[6] Mike Davies, Narayan Srinivasa, Tsung-Han Lin, Gautham Chinya, Yongqiang Cao, Sri Harsha Choday, Georgios Dimou, Prasad Joshi, Nabil Imam, Shweta Jain, et al. Loihi: A neuromorphic manycore processor with on-chip learning. *IEEE Micro*, 38(1):82–99, 2018. 2

[7] Fei Deng, Junyeong Park, and Sungjin Ahn. Facing off world model backbones: RNNs, transformers, and S4. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 1, 2

[8] Maxime Fabre, Lyubov Dudchenko, Younes Bouhadjar, and Emre Neftci. SiLIF: Structured state space model dynamics and parametrization for spiking neural networks. *arXiv preprint arXiv:2506.06374*, 2025. 3

[9] Wei Fang, Zhaofei Yu, Zhaokun Zhou, Ding Chen, Yanqi Chen, Zhengyu Ma, Timothée Masquelier, and Yonghong Tian. Parallel spiking neurons with high efficiency and ability to learn long-term dependencies. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 3

[10] David Ha and Jürgen Schmidhuber. Recurrent world models facilitate policy evolution. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018. 1, 2

[11] Danijar Hafner, Timothy Lillicrap, Ian Fischer, Ruben Villegas, David Ha, Honglak Lee, and James Davidson. Learning latent dynamics for planning from pixels. In *International Conference on Machine Learning (ICML)*, 2019.

[12] Danijar Hafner, Timothy Lillicrap, Jimmy Ba, and Mohammad Norouzi. Dream to control: Learning behaviors by latent imagination. In *International Conference on Learning Representations (ICLR)*, 2020. 1

[13] Danijar Hafner, Timothy Lillicrap, Mohammad Norouzi, and Jimmy Ba. Mastering Atari with discrete world models. In *International Conference on Learning Representations (ICLR)*, 2021.

[14] Danijar Hafner, Jurgis Pasukonis, Jimmy Ba, and Timothy Lillicrap. Mastering diverse control tasks through world models. *Nature*, 640:647–653, 2025. 1, 2, 3, 5, 6, 7, 8

[15] Saya Higuchi, Sebastian Kairat, Sander M. Bohte, and Sebastian Otte. Balanced resonate-and-fire neurons. In *International Conference on Machine Learning (ICML)*, 2024. 3

[16] Mark Horowitz. Computing's energy problem (and what we can do about it). In *IEEE International Solid-State Circuits Conference (ISSCC)*, pages 10–14, 2014. 2, 8

[17] Justus Huebotter, Pablo Lanillos, Marcel van Gerven, and Serge Thill. Spiking neural networks for continuous control via end-to-end model-based learning. *arXiv preprint arXiv:2509.05356*, 2025. 3

[18] Bruce Hutcheon and Yosef Yarom. Resonance, oscillation and the intrinsic frequency preferences of neurons. *Trends in Neurosciences*, 23(5):216–222, 2000. 3

[19] Eugene M. Izhikevich. Resonate-and-fire neurons. *Neural Networks*, 14(6–7):883–894, 2001. 2, 3

[20] Matthew Larkum. A cellular mechanism for cortical associations: An organizing principle for the cerebral cortex. *Trends in Neurosciences*, 36(3):141–151, 2013. 3

[21] Guisong Liu, Wenjie Deng, Xiurui Xie, Li Huang, and Huajin Tang. Human-level control through directly trained deep spiking Q-networks. *IEEE Transactions on Cybernetics*, 53 (11):7187–7198, 2023. 2

[22] Michael London and Michael Häusser. Dendritic computation. *Annual Review of Neuroscience*, 28:503–532, 2005. 3

[23] Chris Lu, Yannick Schroecker, Albert Gu, Emilio Parisotto, Jakob Foerster, Satinder Singh, and Feryal Behbahani. Structured state space models for in-context reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[24] Wolfgang Maass. Networks of spiking neurons: The third generation of neural network models. *Neural Networks*, 10 (9):1659–1671, 1997. 1

[25] Vincent Micheli, Eloi Alonso, and François Fleuret. Transformers are sample-efficient world models. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[26] Steven Morad, Ryan Kortvelesy, Matteo Bettini, Stephan Liwicki, and Amanda Prorok. POPGym: Benchmarking partially observable reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2, 6, 7

[27] Steven Morad, Ryan Kortvelesy, Stephan Liwicki, and Amanda Prorok. Reinforcement learning with fast and forgetful memory. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[28] Emre O. Neftci, Hesham Mostafa, and Friedemann Zenke. Surrogate gradient learning in spiking neural networks: Bringing the power of gradient-based optimization to spiking neural networks. *IEEE Signal Processing Magazine*, 36(6):51–63, 2019. 3

[29] Tianwei Ni, Michel Ma, Benjamin Eysenbach, and Pierre-Luc Bacon. When do transformers shine in RL? decoupling memory from credit assignment. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[30] Garrick Orchard, E. Paxon Frady, Daniel Ben Dayan Rubin, Sophia Sanborn, Sumit Bam Shrestha, Friedrich T. Sommer, and Mike Davies. Efficient neuromorphic signal processing with Loihi 2. In *IEEE Workshop on Signal Processing Systems (SiPS)*, pages 254–259, 2021. 2

[31] Antonio Orvieto, Samuel L. Smith, Albert Gu, Anushan Fernando, Caglar Gulcehre, Razvan Pascanu, and Soham De. Resurrecting recurrent neural networks for long sequences. In *International Conference on Machine Learning (ICML)*, 2023. 2, 4, 5

[32] Ian Osband, Yotam Doron, Matteo Hessel, John Aslanides, Eren Sezener, Andre Saraiva, Katrina McKinney, Tor Lattimore, Csaba Szepesvári, Satinder Singh, Benjamin Van Roy, Richard Sutton, David Silver, and Hado van Hasselt. Behaviour suite for reinforcement learning. In *International Conference on Learning Representations (ICLR)*, 2020. 1, 2, 3, 6, 7

[33] Jurgis Pasukonis, Timothy Lillicrap, and Danijar Hafner. Evaluating long-term memory in 3D mazes. In *International Conference on Learning Representations (ICLR)*, 2023. 1, 2, 6, 8

[34] Panayiota Poirazi, Terrence Brannon, and Bartlett W. Mel. Pyramidal neuron as two-layer neural network. *Neuron*, 37 (6):989–999, 2003. 3

[35] Lang Qin, Ziming Wang, Runhao Jiang, Rui Yan, and Huajin Tang. GRSN: Gated recurrent spiking neurons for POMDPs and MARL. In *AAAI Conference on Artificial Intelligence*, pages 1483–1491, 2025. 2, 3, 6, 7, 8

[36] Jan Robine, Marc Höftmann, Tobias Uelwer, and Stefan Harmeling. Transformer-based world models are happy with 100k interactions. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[37] Kaushik Roy, Akhilesh Jaiswal, and Priyadarshini Panda. Towards spike-based machine intelligence with neuromorphic computing. *Nature*, 575:607–617, 2019. 1

[38] Mohammad Reza Samsami, Artem Zholus, Janarthanan Rajendran, and Sarath Chandar. Mastering memory tasks with world models. In *International Conference on Learning Representations (ICLR)*, 2024. 1, 2, 6, 7

[39] Shuaijie Shen, Chao Wang, Renzhuo Huang, Yan Zhong, Qinghai Guo, Zhichao Lu, Jianguo Zhang, and Luziwei Leng. SpikingSSMs: Learning long sequences with sparse and parallel spiking state space models. In *AAAI Conference on Artificial Intelligence*, pages 20380–20388, 2025. 3

[40] Matei-Ioan Stan and Oliver Rhodes. Learning long sequences in spiking neural networks. *Scientific Reports*, 14:21957, 2024. 3

[41] Yinqian Sun, Feifei Zhao, Mingyang Lyu, and Yi Zeng. Spiking world model with multicompartment neurons for model-based reinforcement learning. *Proceedings of the National Academy of Sciences*, 122(50):e2513319122, 2025. 2, 3, 6, 7, 8

[42] Guangzhi Tang, Neelesh Kumar, Raymond Yoo, and Konstantinos P. Michmizos. Deep reinforcement learning with population-coded spiking neural network for continuous control. In *Conference on Robot Learning (CoRL)*, 2020. 2

[43] Yuval Tassa, Yotam Doron, Alistair Muldal, Tom Erez, Yazhe Li, Diego de Las Casas, David Budden, Abbas Abdolmaleki, Josh Merel, Andrew Lefrancq, Timothy Lillicrap, and Martin Riedmiller. DeepMind control suite. *arXiv preprint arXiv:1801.00690*, 2018. 6

[44] Aaron Voelker, Ivana Kajić, and Chris Eliasmith. Legendre memory units: Continuous-time representation in recurrent neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019. 2

[45] Erik Wijmans, Manolis Savva, Irfan Essa, Stefan Lee, Ari S. Morcos, and Dhruv Batra. Emergence of maps in the memories of blind navigation agents. In *International Conference on Learning Representations (ICLR)*, 2023. 2

[46] Yujie Wu, Lei Deng, Guoqi Li, Jun Zhu, and Luping Shi. Spatio-temporal backpropagation for training high-performance spiking neural networks. *Frontiers in Neuroscience*, 12:331, 2018. 3

[47] Bojian Yin, Federico Corradi, and Sander M. Bohté. Accurate and efficient time-domain classification with adaptive spiking recurrent neural networks. *Nature Machine Intelligence*, 3 (10):905–913, 2021. 2, 3

[48] Dehao Zhang, Malu Zhang, Shuai Wang, Jingya Wang, Wenjie Wei, Zeyu Ma, Guoqing Wang, Yang Yang, and Haizhou Li. Dendritic resonate-and-fire neuron for effective and efficient long sequence modeling. *arXiv preprint arXiv:2509.17186*, 2025. 3

[49] Shimin Zhang, Qu Yang, Chenxiang Ma, Jibin Wu, Haizhou Li, and Kay Chen Tan. TC-LIF: A two-compartment spiking neuron model for long-term sequential modelling. In *AAAI Conference on Artificial Intelligence*, pages 16838–16847, 2024. 2, 3, 6, 7, 8

[50] Weipu Zhang, Gang Wang, Jian Sun, Yetian Yuan, and Gao Huang. STORM: Efficient stochastic transformer based world models for reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023. 2

[51] Hanle Zheng, Zhong Zheng, Rui Hu, Bo Xiao, Yujie Wu, Fangwen Yu, Xue Liu, Guoqi Li, and Lei Deng. Temporal dendritic heterogeneity incorporated with spiking neural networks for learning multi-timescale dynamics. *Nature Communications*, 15:277, 2024. 2, 3, 6, 7, 8